

3 In this question you must show detailed reasoning.

The roots of the equation $x^2 - 2x + 4 = 0$ are α and β .

(a) Find α and β in modulus-argument form. **[4]**

(b) Hence or otherwise show that α and β are both roots of $x^3 + \lambda = 0$, where λ is a real constant to be determined. **[3]**